

Networking Concepts

Routing, VLAN, and TCP/UDP — Easy Study Notes

Reorganized from Lecture 10 for simple and complete reading

What This Note Covers

- 1. Routing — what a router does and how it picks a path
- 2. Routing Protocols — static vs dynamic, and common protocol names
- 3. Internetworking Devices — hub, bridge, switch, router
- 4. VLAN — virtual networks and their benefits
- 5. TCP/UDP — the two main transport protocols

Note: Each topic is short and simple on purpose. Read it once, and you will cover everything in the original slides.

1. Routing

1.1 What Is a Router?

A router is a device. Its job is to decide the next point where a data packet should go, so the packet can reach its final destination.

- A router lets different networks talk to each other.
- A router keeps a table of all the routes it knows, along with their condition. It uses this table to pick the best route for each packet.
- A packet usually passes through several routers before it reaches its destination.
- If there are many possible routes, the router picks the one with the lowest weight (also called metric) first.

Note: Think of a router like a traffic signal for data. It always looks for the shortest and safest path.

1.2 How Routing Looks in a Network

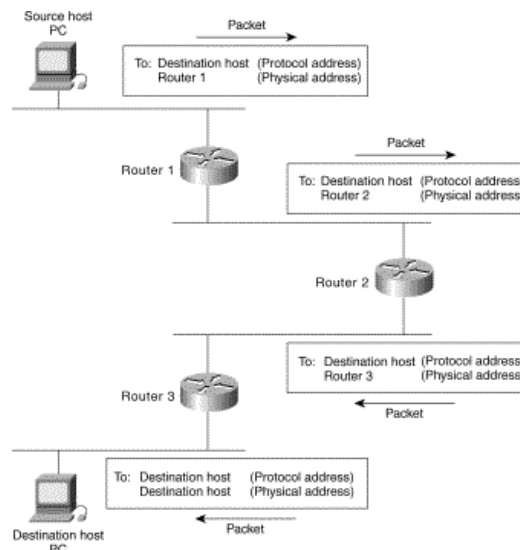


Figure 1: A simple picture of how routing connects different networks together.

1.3 Routing Protocols

A routing protocol is a set of rules that routers use to share route information with each other. There are two main types.

- Static Routing — the route is set by hand. It does not change on its own.
- Dynamic Routing — routers learn routes automatically and update them when the network changes.

Dynamic routing protocols are further grouped by where they work:

- IGP (Interior Gateway Protocol) — sends route data inside one autonomous system (one organization's network). Examples:
 - RIP (Routing Information Protocol)

- RIP-2 (RIP Version 2)
- OSPF (Open Shortest Path First)
- IGRP (Interior Gateway Routing Protocol)
- EIGRP (Enhanced Interior Gateway Routing Protocol)
- IS-IS
- EGP (Exterior Gateway Protocol) — sends route data between different autonomous systems. Example:
 - BGP (Border Gateway Protocol)

Note: An autonomous system just means one network that is managed by one organization, like a company or an internet service provider.

2. Internetworking Devices

These are the main devices used to build and connect networks. Each one works at a different level and has a different job.

Device	What It Does
Hub	Connects many devices to one network. It repeats and resends the signal to all ports, without checking where it should go.
Bridge	Splits one network into smaller parts. It works at Layer 2 (data link layer) and does not depend on higher-level protocols.
Switch	Works like a bridge but has more ports. Each port is its own separate segment, so it reduces data collisions and improves speed.
Router	Connects different networks and separates broadcast traffic. It sends data based on the network address (Layer 3), not the device's hardware address.

Note: Simple way to remember: Hub = repeats to everyone. Bridge/Switch = splits traffic smartly. Router = connects separate networks.

3. VLAN (Virtual LAN)

3.1 What Is a VLAN?

A VLAN lets network managers group users by logic, such as their job, project, or the application they use, instead of grouping them by physical location.

- Data can only move between VLANs through routing.
- VLANs do the same job that physical routers used to do — they divide a LAN into smaller, separate parts.

3.2 VLAN and Inter-VLAN Routing

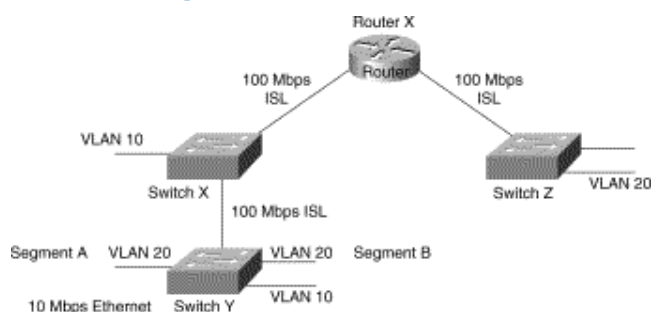


Figure 2: VLANs are separate groups. A router is needed for them to communicate with each other.

3.3 Why Use VLANs?

- Broadcast Control — Just like switches keep collision traffic separate, VLANs keep broadcast and multicast traffic inside their own group.
- Security — If routing is turned off for a VLAN, users outside that VLAN cannot reach users inside it, and the other way around. This gives strong protection for sensitive projects.
- Performance — Users who need faster or separate networking can be placed in their own VLAN.

Note: VLANs are useful because they give the benefits of separate physical networks, without needing separate physical cables.

4. TCP / UDP

TCP and UDP are both Transport Layer protocols. This layer is responsible for moving data between two devices.

Feature	TCP	UDP
Connection Type	Connection-oriented (a link is set up before sending data)	Connectionless (data is sent with no set-up)
Reliability	Uses checksum and acknowledgment to check and confirm delivery	No delivery check; faster but less reliable
Port Number	Uses a 16-bit source and destination port number	Uses a 16-bit source and destination port number

4.1 Standard Port Numbers

Common applications always use the same port number, so devices know which service to connect to.

- Email — Port 25
- Telnet — Port 23
- FTP — Ports 20 and 21
- SSH — Port 22

Note: TCP is used when accuracy matters, like sending a file. UDP is used when speed matters more than perfect accuracy, like video calls.

Quick Summary

- Routing moves packets across networks using the best path.
- Routing protocols can be static (fixed) or dynamic (self-updating), and are grouped as IGP or EGP.
- Hubs, bridges, switches, and routers each connect networks in a different way.
- VLANs group users logically for better control, security, and performance.
- TCP is reliable and connection-based. UDP is fast and connectionless.

Note: Reading this summary after the full notes is a good way to check what you remember.